

Speech Large Language Models in the Real World: A Survey of Training, Evaluation, Deployment, and Responsible Design

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ABSTRACT

Speech Large Language Models (Speech LLMs) extend text-based language models to spoken interaction, preserving linguistic content together with paralinguistic cues such as emotion, prosody, and speaker identity. Existing surveys describe how Speech LLMs are built; this survey instead asks what evidence and system conditions are required before Speech LLMs can be evaluated, deployed, and governed as real-world interactive systems. We adopt a *lifecycle* framing connecting four stages—training, evaluation, deployment, and responsible design—and argue that progress in any one stage is constrained by choices in the others. The survey makes four contributions. First, we synthesize training strategies for modality alignment, weakly supervised pretraining, parameter-efficient fine-tuning, and preference alignment via RLHF, DPO, and RLAI. Second, we conduct a meta-evaluation of recent Speech LLM benchmarks, analyzing construct, ecological, evaluator, and operational validity rather than only capability coverage. Third, we synthesize deployment research around real-time interaction constraints—streaming, serving, edge feasibility, and full-duplex behavior. Fourth, we develop a speech-specific responsible-design framework covering audio jailbreaks, voice cloning, speaker privacy, and provenance. By treating Speech LLMs as interactive systems rather than isolated models, the survey identifies the structural couplings that future research must address.

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Shiqi Yang: Conceptualization, Methodology, Investigation, Writing – Original Draft, Writing – Review & Editing, Visualization. **Ziyi Huang:** Investigation, Writing – Original Draft, Writing – Review & Editing. **Wengran Xiao:** Investigation, Writing – Original Draft, Writing – Review & Editing.

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